CS23431 - OPERATING SYSTEM

EXP 2: SJF Using C

QUESTION :

Write a C program to implement the Non-Preemptive Shortest Job First (SJF) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), and Burst Times (BT). At any given time, the process with the minimum burst time among the arrived processes must be executed first. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and find the overall average TAT and WT.

CODE:

```
#include <stdio.h>

int main() {
int n, i, completed = 0, time = 0, min_index;
printf("Enter number of processes: \n");
scanf("%d", &n);
int AT[n], BT[n], CT[n], TAT[n], WT[n], done[n];
for(i = 0; i < n; i++) {
    printf("Enter Arrival Time and Burst Time for P%d: \n", i + 1);
    scanf("%d %d", &AT[i], &BT[i]);
    done[i] = 0;
}
while(completed < n) {
    min_index = -1;
    for(i = 0; i < n; i++) {
        if(!done[i] && AT[i] <= time) {
            if(min_index == -1 || BT[i] < BT[min_index])
                min_index = i;
        }
    }
    if(min_index == -1) {
        time++;
    } else {
        time += BT[min_index];
        CT[min_index] = time;
        done[min_index] = 1;
        completed++;
    }
}
```

```
    float totalTAT = 0, totalWT = 0;
    for(i = 0; i < n; i++) {
        TAT[i] = CT[i] - AT[i];
        WT[i] = TAT[i] - BT[i];
        totalTAT += TAT[i];
        totalWT += WT[i];
    }
    printf("\nAverage Turnaround Time = %.2f\n", totalTAT / n);
}
```

Output:

Custom Test — Success

Input :

```
4
0 6
0 8
0 7
0 3
```

Expected Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:
Enter Arrival Time and Burst Time for P4:

Average Turnaround Time = 13.00
Average Waiting Time = 7.00
```

Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:
Enter Arrival Time and Burst Time for P4:

Average Turnaround Time = 13.00
Average Waiting Time = 7.00
```